Developing Applications on Qt

In the previous article in this series we went over the installation and some basic examples of Qt. In this article, we will learn how to use the Qt classes for basic data-types and lists.

Part—2

Let's begin with the basic data-type class, i.e., *Qchar,* which is based on 16-bit Unicode characters, which are 16-bit unsigned data. In C we have *'char'* single-byte variables limited to expressing 255 characters, but with 16 bits, many more characters can be used. Some important member functions of this class are: *isdigit(), isLetter(), isLower(), isUpper(),isSymbol(), toLower(), toUpper()*, etc. To use this class, you need to include the *QtCore* library. Let's look at an example. As discussed in the previous article, create a new project in Qt Creator. From the list of project type options, I have chosen Qt Console Application, since I'm not using any graphic widgets. In the *main.cpp* file, enter the following code:

```
#include <QtCore/QCoreApplication>
#include <QtCore>
int main(int argc, char *argv[])
{
    QCoreApplication a(argc, argv);
    QChar ch;
    ch = 'M';
    if(ch.isLetter())
    {
        if(ch.isLower())
            qDebug() << "Ch is a Lower Letter";
        else
            qDebug() << "Ch is an Upper Letter";
    }
    qDebug() << sizeof(QChar);
    return a.exec();
}
```

Press *Ctrl+r* to build and execute the project. A pop-up window will display the following output:

```
Ch is an Upper Letter
2
```

To master the more than 20 member functions of the *QChar* class, write small programs to test them out, one by one.

The next important class is *QString*. Most times, project work requires dealing with strings. Qt has a very handy class called *QString* (a string of Unicode characters) whose base element is *QChar*. Qt provides another option to store data in string-like form—*QByteArray*, which stores raw bytes. But to make your application support multiple languages, use *QString* for user-interface strings. The following example uses some of the important member functions:

```
#include <QtCore/QCoreApplication>
#include <QtCore>
int main(int argc, char *argv[])
{
    QCoreApplication a(argc, argv);
    QString str("Linux");
    //QString str = "Linux";
    qDebug() << str.size() << str;
    str.append(" Kernel");
    qDebug() << str.size() << str;
    str.insert(5,"3.0");
    qDebug() << str.size() << str;
    if(str.startsWith("Linux"))
    {
        qDebug() << "The string starts with Linux";
    }
    int ret = QString::compare(str, "Linux3.0 Kernel",
Qt::CaseInsensitive);
    if(!ret)
    {
        qDebug() << "Strings are equal";
    }
    return a.exec();
}
```

Running it yields the following output:

```
5 "Linux"
12 "Linux Kernel"
15 "Linux3.0 Kernel"
The string starts with Linux
Strings are equal
```